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Beyond Pixels and Words : Frames of Reference and Priming in Vision Language Models

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Abstract

This thesis investigates how Vision Language Models (VLMs) interpret spatial relations using different Frames of Reference (FoR). We examine four questions: which FoR models prefer, whether they can distinguish between relative and intrinsic FoRs, whether dialogue priming influences FoR choice, and whether landmark type affects this choice. A controlled 3D dataset is generated using Python scripts in Blender, with scene annotations and questions designed specifically for the experiments. Two MLLMs are evaluated using controlled 3D scenes in three experiments. The results show that both models strongly prefer the speaker FoR, while models struggle with the intrinsic FoR. This preference also persists across dialogue turns despite intrinsic FoR priming. Moreover, we found that responses were significantly less likely to use the speaker FoR when the landmark was a flamingo compared with a sofa or a bed. Overall, our research shows that current MLLMs have a strong speaker bias and may have limited flexibility in interpreting spatial relations from different perspectives.

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1 Introduction

Expressions such as *right of*, *left of*, and *in front of* are commonly used to describe the spatial location of the objects in our daily life. However, the spatial relationships are not fixed but determined by several factors. Based on Levinson, 2003, spatial relations can be established based on different coordinate systems, including the observer's viewpoint (relative FoR), the intrinsic orientation of the reference object (intrinsic FoR), or fixed environmental directions (absolute FoR).

As vision language models (VLMs) have rapidly advanced in recent years, they have shown impressive capabilities in spatial reasoning. However, most existing VLMs are primarily trained on datasets that spatial information is often observed from a single viewpoint. For instance, the classic CLEVR dataset (Johnson et al., 2017) generates 3D scenes from a single fixed camera viewpoint, where spatial relations are described based on the rendered visual perspective. Such dataset limitations may restrict models' ability to perform spatial reasoning under different frames of reference.

There are several recent studies related to spatial reasoning in different FoR. Liu et al. (Liu, Emerson, and Collier, 2023) introduce the Visual Spatial Reasoning (VSR) dataset, which evaluates whether VLMs can understand spatial relations in images. The dataset contains spatial expressions involving different reference perspectives, including intrinsic and viewpoint interpretations. Similarly, Lee et al. (Lee et al., 2022) create a dataset of visual questions involving spatial relations, where some questions using an intrinsic FoR and others using the camera viewpoint.

A Human study further shows that spatial perspectives are flexible rather than fixed. Dobnik et al. (Dobnik, Koniaris, and Kelleher, 2013) investigate the role of priming and perspective alignment in dialogue, showing that speakers may adapt their choice of spatial reference frames to the context and maintain perspective alignment across several turns, but only locally, until contextual factors change.

Inspired by these findings, we decide to investigate the ability of VLMs to reason about spatial relations in different kinds of FoRs and further investigate whether their FoR preference can be influenced through priming.

To achieve our goals, we developed a Python script to automatically generate 3D designed scenes in Blender, with their corresponding annotations and questions. Each scene contains objects with clearly defined intrinsic orientations, such as sofas, chairs, tables, and lamps. Each image is associated with three experimental rounds, each consisting of a different set of questions designed for a specific experimental purpose and to

address one or two corresponding research question. More detailed information about the scene generation and dataset will be provided in the Experiments chapter.

2 background

In this chapter, we discuss FoRs in detail and also provide further background on priming.

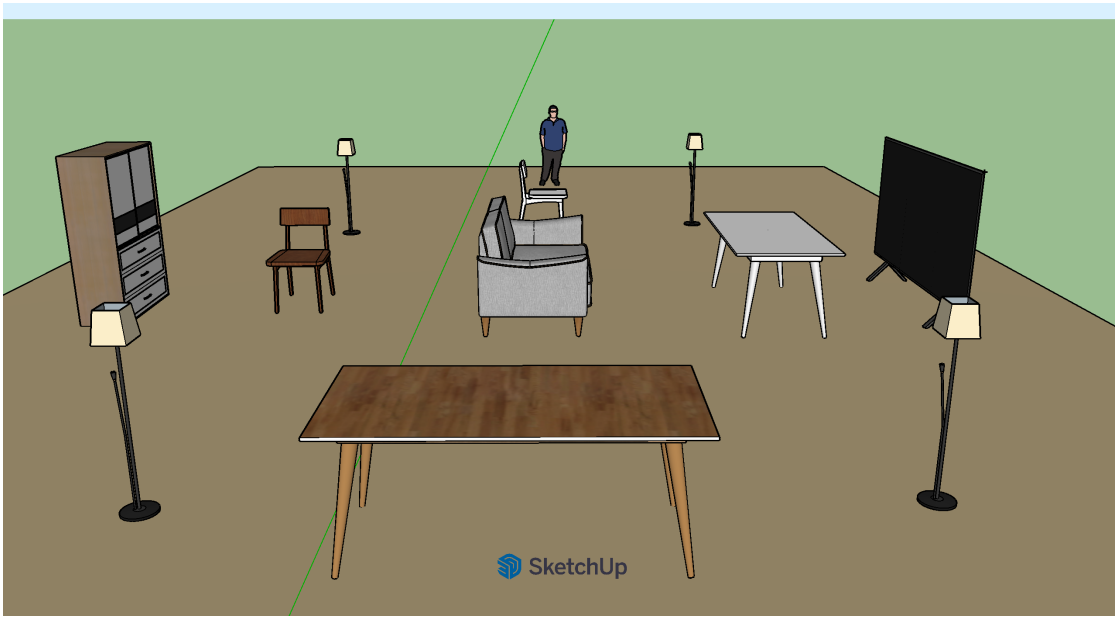


Figure 2.1: Example of a 3D scene generated in SketchUp.

2.1 Frame of reference

As we have mentioned, there is no fixed way for humans to describe spatial relations between two or more objects. To comprehensively describe spatial relations, we need a system that possibly includes all of these relations. And FoR is such a system. In spatial language, even when the same objects are in the same positions, the description can still be different. For instance, in Figure 2.1, we can describe the position of the white table in several ways: the table is to the left of the sofa, the table is in front of the TV, and the table is in the right area of the house. In fact, we are not just talking about one single object, but determining its position by choosing different reference objects. And obviously, some of the objects have their own orientation.

2.1.1 Relative

Relative FoR means that we choose the observer's perspective. In this case, the man wearing a blue T-shirt serves as the hearer. Therefore, the white table can be described as being to the right of the sofa from the speaker's perspective, or to the left of the sofa from the hearer's perspective.

2.1.2 Intrinsic

Unlike the relative frame of reference, the intrinsic frame of reference is determined by the inherent orientation of the reference object. Therefore, we can say that the white table is in front of the sofa based on its intrinsic orientation, while at the same time, the white table can be described as being to the right of the sofa from the speaker's perspective.

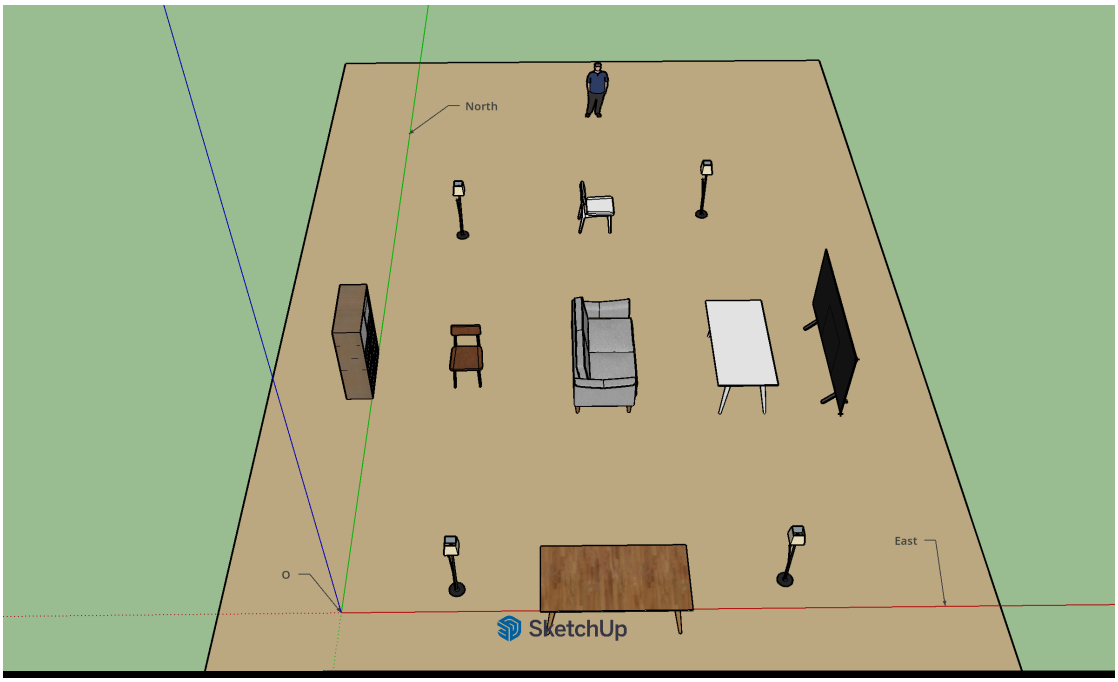


Figure 2.2: Example of a 3D scene generated in SketchUp with coordinate system.

2.1.3 Absolute

Absolute FoR does not use the observer's viewpoint or the intrinsic orientation of the reference object. Instead, spatial relations are determined by an external, fixed coordinate system. As we can see in Figure 2.2, in this case, we can say that the white table is to the east of the sofa.

2.2 Spatial FoR Priming in Humans

An interesting research we mentioned above has found that humans may adopt the FoR used by their interlocutors when describing spatial relations. This phenomenon is generally referred to as spatial language priming. In other words, the way in which a person describes a spatial relation can influence how another person describes subsequent spatial relations. For instance, if a speaker repeatedly uses an intrinsic FoR to describe spatial relations, the hearer is also more likely to switch to an intrinsic FoR when describing subsequent spatial relations.

3 Experiemnt

3.1 Experiment Design

We design three tasks here to answer several different question: "What is the most common persepctive they will use in the answer?" , "Can VLLM understand difference between two kinds of FoR?(relative and intrinsic)" , "Does dialogue priming influence the FoR?" and "Does salience of the landmark affect the choice?".

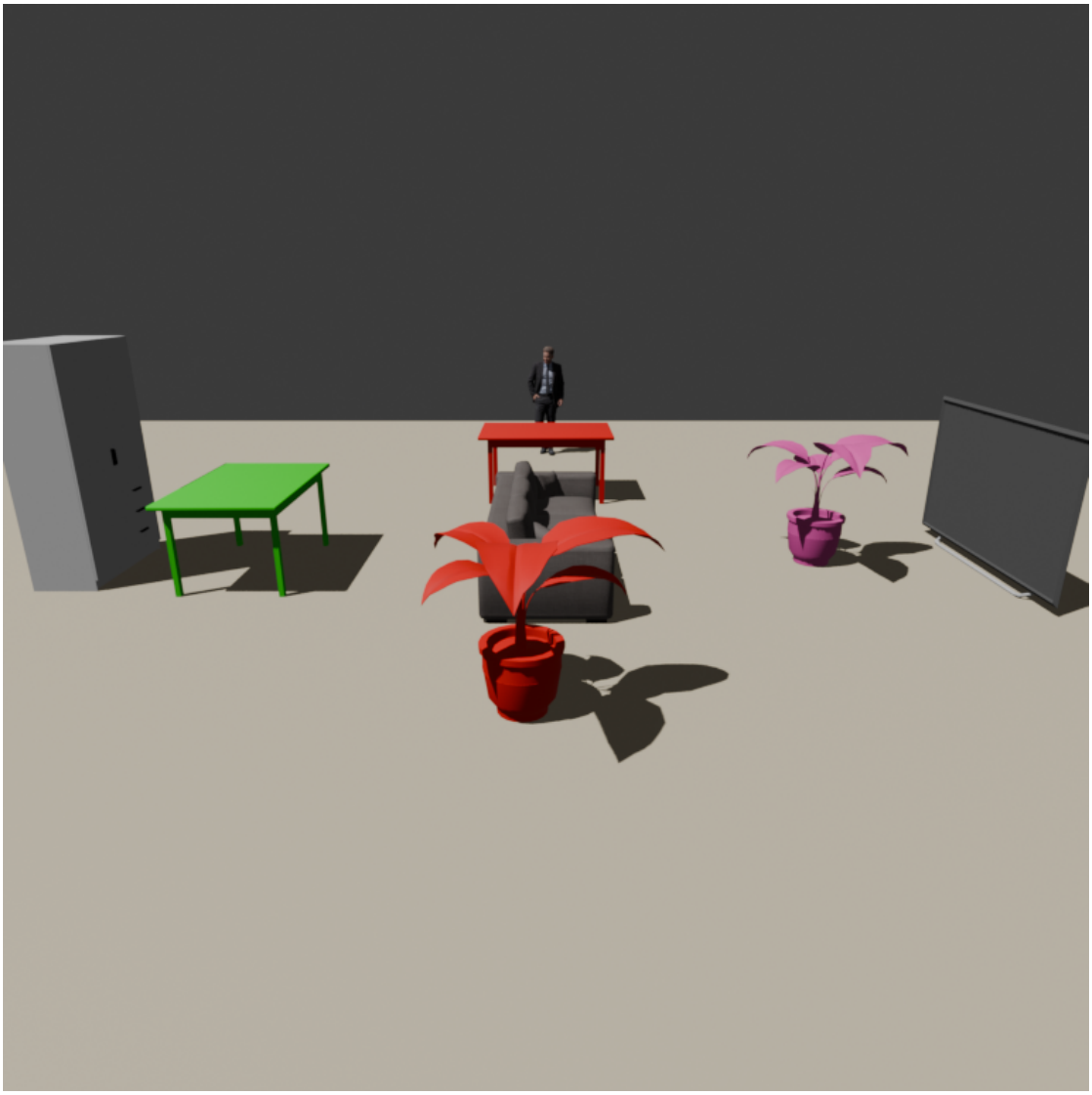


Figure 3.1: Example of a 3D scene generated in blender.

Firstly, we give the model maximum freedom to answer simple spatial questions, such as: "Where is the red table?" Each answer will be recorded and used for analysis in the next step. This procedure constitutes Experiment 1. Then, the next task examines whether MLLMs can distinguish different FoR. We keep the scene constant and manipulate only the prompt. The model is asked the same spatial question three times about the same image from three different perspective, each time under a different instruction. It is worth noting here that the models are asked to answer questions using a specific FoR, which constitutes a form of priming. For instance, as shown in 3.1, in the first condition, the question is grounded in the speaker perspective (e.g. "From your perspective looking at the scene, what is to the left of the sofa?"). In the second condition,

it is grounded in the intrinsic FoR of an object (e.g. "From the sofa's own perspective, what is to the left of the sofa?"). In the third condition, the model is asked to adopt a hearer's perspective by being told that the interlocutor is standing opposite, facing the scene from 180° (e.g. "I am standing opposite you, facing you. From my perspective, what is to the left of the sofa?"). This procedure constitutes Experiment 2.

The third task is designed to solve the rest questions. In this task, each model is presented with a scene and a four-turn vision question answering sequence. In Turn 1, a spatial description grounded in the intrinsic FoR is provided as context (e.g. There is a TV in front of the sofa), implicitly establishing the object's intrinsic orientation as the reference frame. In Turns 2 and 3, the model is asked an ambiguous spatial question whose answer differs depending on which FoR is applied. (e.g. Is there a table to the left of the sofa? In this case, the tested model decides which FoR to use. It's worth noting that both different colour tables satisfy the condition, but under different FoRs. To distinguish which object the model is referring to, the two objects of the same type are assigned different colors. The model is additionally required to specify the color of the object it refers to. In Turn 4, the model is asked to freely describe the location of an object : Where is the wardrobe? Please describe its location. The generated description is analysed to determine which FoR the model adopts, and whether this choice reflects the frame established in Turn 1 or defaults to an alternative frame such as the relative FoR. It is important to note that alignment with the previously established FoR is fundamentally different from accuracy. Alignment does not indicate whether an answer is "correct"; instead, it captures the model's tendency to maintain or adopt the same FoR across dialogue turns. Consequently, a higher alignment rate is not inherently better. This procedure constitutes Experiment 3. It is worth noting that all objects will be varied across scenes, except for the TV and wardrobe, allowing us to investigate whether landmark type affects the priming effect.

3.2 Dataset

3.2.1 Dataset Generation

We generate the scenes using a Python script in blender, and detailed information about each object's properties, position, and spatial relations to the landmark is recorded as annotations. These annotations are then used to generate questions in subsequent steps. Each scene contains 11 questions: four open-ended questions (e.g., "Where is the red table?" referring to an ordinary object), three priming questions (e.g., "From the speaker's perspective, the sofa's intrinsic perspective, or the hearer's perspective, where is the table?"), and a four-turn dialogue consisting of four questions.

3.2.2 Scene Composition

A tested scene includes one landmark (e.g., a sofa, bed, or flamingo), one priming object (a TV or wardrobe), one evaluation object (a TV or wardrobe, depending on the priming object), four ordinary objects (a table, chair, lamp, or plant), and one hearer. There

are several constraints on the scenes that should be noted. First, the landmark, which is located at the center of the scene, should have a clear orientation, allowing spatial relations to be interpreted using an intrinsic FoR. Meanwhile, the priming object and the evaluation object should be of different types, which means they can not both be TVs or wardrobes. The landmark, priming object, and evaluation object have no color information, and their default colors are used. There are four ordinary objects in total, all of which have intrinsic orientations and color features.

3.2.3 Object Position Constraints

The positions and orientations of the objects are strictly controlled to create situations in which different FoRs may lead to different interpretations. The landmark is located at the center of the scene, and its orientation is restricted to either the positive or negative direction of the x-axis, corresponding to the right or left side from the speaker's perspective. The priming and evaluation objects always face the landmark and are distributed on opposite sides of the landmark along the x-axis, outside the positions occupied by the ordinary objects. The four ordinary objects must also satisfy specific constraints. First, the two objects located to the right of the landmark from the speaker's perspective and to the right of the landmark from its intrinsic perspective must be of the same type but different colors. This design is used to determine which object the model refers to. The same constraint applies to the two objects on the corresponding left sides. Finally, the hearer is positioned directly opposite the speaker.

3.3 Models

Two models are tested in our experiments:Gemini-3.1-Flash-Lite and Kimi 2.6. Both Gemini-3.1-Flash-LiteGoogle, 2026 and Kimi 2.6Moonshot AI, 2026 are accessed through APIs. Gemini 3.1 Flash-Lite is a multimodal model from Google. It is a low-latency, cost-effective multimodal model optimized for high-frequency, lightweight tasks. The model supports text, image, video, audio, and PDF inputs, and is designed for high-volume agentic workflows, simple data extraction, and applications where latency and API cost are the primary constraints. Kimi K2.6 is a multimodal model developed by Moonshot AI. It supports text, image, and video inputs, as well as both thinking and non-thinking modes, and provides a 256K-token context window. In this study, Kimi K2.6 is accessed through the Kimi API. Further details about the models can be found in the official documentation.

3.4 Evaluation

We used regular expression matching to identify predefined spatial relations in the model responses and classify them accordingly. The experimental outcomes are then evaluated mainly using quantitative measures, including answer accuracy and FoR distributions. For Experiment 1, which investigates "What is the most common perspective they will

use in the answer?", we focus on answer accuracy to evaluate the models' performance. The same evaluation approach is applied to Experiment 2. For Experiment 3, we analyse the FoRs used in the model responses and count the FoRs selected by the models for each landmark type allowing us to address the final research question, "Does salience of the landmark affect the choice?"

3.5 Ethical Considerations

As we mentioned, all of the experimental scenes are generated using Blender and therefore do not contain identifiable individuals or private information. External VLMs are only provided with these synthetic images and corresponding research questions.

4 Results

4.1 Experiment 1

Table 4.1: Frame of Reference (FoR) distribution in Experiment 1.

Model	Speaker	Intrinsic	Hearer	Ambiguous	Unknown	Total
Gemini-3.1-Flash-Lite	233 (58.25%)	7 (1.75%)	38 (9.50%)	72 (18.00%)	50 (12.50%)	400
Kimi-K2.6	242 (60.50%)	3 (0.75%)	31 (7.75%)	30 (7.50%)	94 (23.50%)	400

Table 4.1 shows the results of Experiment 1, displaying the FoRs that the models chose when answering the spatial questions. The "Ambiguous" label refers to cases where the model uses at least two different perspectives to answer a question, while "Unknown" refers to cases where the model uses spatial expressions that are not fully unambiguous, such as "near" and "between".

As shown in Table 4.1, both models tend to use the speaker FoR most frequently, accounting for approximately 60% of all responses. Excluding the "Ambiguous" and "Unknown" categories, the hearer FoR is the second most frequent category, although its frequency decreases substantially compared with the speaker FoR (Gemini decreases from 58.25% to 9.50% , and Kimi decreases from 60.50% to 7.75%). In contrast, the intrinsic FoR is rarely used, accounting for only 1.75% of Gemini's responses and 0.75% of Kimi's responses.

4.2 Experiment 2

Table 4.2: Accuracy by Frame of Reference (FoR) in Experiment 2.

Model	Speaker	Intrinsic	Hearer	Overall
Gemini-3.1-Flash-Lite	70/100 (70.00%)	2/100 (2.00%)	46/100 (46.00%)	39.33%
Kimi-K2.6	66/100 (66.00%)	5/100 (5.00%)	3/100 (3.00%)	24.67%

Table 4.2 shows the results of Experiment 2. As shown in the table, both models achieve relatively good performance on the speaker FoR, with accuracies of 70% and 66% . For the hearer FoR, Gemini still achieves a moderate accuracy of 46%, whereas Kimi's

performance drops substantially. Meanwhile, the intrinsic FoR appears to be the most difficult FoR for both models to understand, with Gemini and Kimi achieving accuracies of only 2% and 5% respectively.

4.3 Experiment 3

4.3.1 Overall Results

Table 4.3: Overall FoR distribution in Experiment 3.

	Gemini			Kimi		
	T2	T3	T4	T2	T3	T4
Speaker	94	94	100	92	88	100
Intrinsic	0	0	0	0	0	0
Unknown	6	6	0	8	12	0
Total	100	100	100	100	100	100

Table 4.3 shows the results of Experiment 3, which consists of a four-turn dialogue. As shown in the results, both models strongly adhere to the speaker's perspective, with almost all responses using the speaker FoR.

4.3.2 Results by Landmark

Table 4.4: FoR distribution in Experiment 3 by landmark.

Model	Landmark	T2			T3			T4	
		S	I	U	S	I	U	S	I
Gemini	Sofa	35	0	2	35	0	2	37	0
Gemini	Flamingo	22	0	2	23	0	1	24	0
Gemini	Bed	37	0	2	36	0	3	39	0
Kimi	Sofa	35	0	2	32	0	5	37	0
Kimi	Flamingo	20	0	4	20	0	4	24	0
Kimi	Bed	37	0	2	36	0	3	39	0

Note. S = Speaker FoR; I = Intrinsic FoR; U = Unknown.

Table 4.4 shows the results of Experiment 3 broken down by landmark type. As mentioned above, both models strongly adhere to the speaker's perspective. However, we observe an interesting phenomenon that we observe more "unknown" responses when the landmark is a flamingo. These responses often contain relatively vague spatial expressions, such as "between," "near," or "close to," making it difficult to reliably identify

the underlying Frame of Reference (FoR). This pattern suggests that the type of landmark may influence how models describe spatial relations and, consequently, their choice of FoR.

5 Conclusion

This thesis mainly investigated four questions: "What is the most common perspective models use in their answers?", "Can VLLMs distinguish between relative and intrinsic FoRs?", "Does dialogue priming influence FoR choice?" And "Does landmark salience affect this choice?".

Experiment 1 shows that both Gemini and Kimi tend to use speaker FoR, and the propotion is much higher than any other FoRs. Experiment 2 further shows that both models can get good accuracy when the priming is speaker FoR. When it comes to hearer FoR, their performance decreases under hearer FoR. Gemini maintains relatively good accuracy, whereas Kimi's performance decreases substantially. Both models also struggle with intrinsic FoR, even under explicit priming.

Experiment 3 shows that this preference remained strong across dialogue turns, even when an intrinsic FoR was introduced in the preceding context. This suggests that, unlike humans, the models did not consistently align their FoR choices with those of their dialogue partner. But we have to note here: this alignment should be understood as a tendency rather than an accuracy measure: a higher alignment rate does not necessarily indicate better performance. Interestingly, We found that flamingo scenes resulted in more unknown responses than sofa and bed scenes, suggesting that landmark type may influence how models describe spatial relations.

Overall, the results suggest that current VLMs have a strong speaker FoR bias in spatial reasoning and do not treat different FoRs equally. This can be seen as a limitation in their ability to flexibly interpret spatial relations from different perspectives.

6 Future work

One of the most challenging aspects of this thesis was the evaluation of the results. Since the models were given considerable freedom in their responses, they often did not follow a consistent format, making the analysis lengthy and complex. Despite the effort to systematically classify the responses, some answers still could not be reliably assigned to a specific category.

In addition, the scenes could be further expanded in future work. The current dataset contains three types of landmarks, two types of special objects (priming and evaluation objects), and four types of ordinary objects. A larger collection of 3D models could be incorporated to create more diverse and complex scenes, allowing future studies to examine whether the observed patterns generalize across a wider range of visual contexts.

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